Membership Inference and Model Leakage: Evaluating Regularization and DP-SGD Defenses on Sensitive Clinical Records

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Abstract

Machine learning models trained on sensitive electronic health records (EHR) and genomic biomarker cohorts risk leaking private patient participation through **Membership Inference Attacks (MIAs)**. Determining that a specific patient's record was included in a disease-specific training cohort directly discloses confidential medical diagnoses. In this technical report, we design and execute an end-to-end empirical investigation of membership leakage on the **CliniLeak** benchmark ($N = 2,400$ clinical and genomic records, $d = 28$ biomarkers, $C = 6$ phenotypes). We implement three black-box attack methodologies—Loss-Threshold (Yeom et al.), Confidence-Entropy (Salem et al.), and Supervised Shadow-Model Classification (Shokri et al.)—and evaluate five defense configurations spanning L_2 weight regularization, Inverted Dropout with Confidence Smoothing, and **Differentially Private Stochastic Gradient Descent (DP-SGD)** with Rényi Differential Privacy (RDP) accounting. Our results demonstrate that an unprotected Multi-Layer Perceptron suffers severe memorization (39.50% generalization gap), yielding an overall Shadow MIA AUC of **0.7390** and an alarming **0.8832 AUC** on rare clinical subpopulations. While heuristic regularizers (L_2 and Dropout+Smoothing) improve test accuracy by +3.00 to +4.17 percentage points with negligible runtime cost and reduce overall MIA AUC to 0.5839, rare patients remain vulnerable (0.6778 AUC). Only **DP-SGD** ($\epsilon = 2.86, \delta = 10^{-5}$) bounds worst-case per-record sensitivity, reducing overall MIA AUC to **0.5445** and rare subgroup disparity ΔAUC from +0.1789 down to +0.0250, at a quantified cost of 8.83 percentage points in test accuracy and a 1.33× training runtime overhead.

1 Introduction and Case Study Motivation

Modern clinical decision support systems increasingly rely on neural networks trained over multi-modal patient registries combining vital signs, metabolic panels, and genomic loci. However, overparameterized neural networks frequently memorize idiosyncratic features of their training examples [1, 5]. In a **Membership Inference Attack (MIA)**, an external adversary with black-box query access to a trained model f_{θ^*} attempts to infer whether a candidate patient record $z = (x, y)$ was present in the private training dataset $D^{\text{in}}_{\text{target}}$.

In healthcare settings, membership inference constitutes a direct privacy breach under HIPAA, GDPR Article 9, and India's Digital Personal Data Protection (DPDP) Act 2023. Knowing that an individual was part of a hospital's training cohort for early-stage oncology or rare neuro-genetic syndromes immediately reveals their diagnosis. Furthermore, privacy leakage is rarely uniform across demographics: patients with rare or atypical clinical presentations induce larger gradient updates during training, rendering them disproportionately vulnerable to memorization.

Contributions:

1. We build a reproducible evaluation pipeline (`run_pipeline.py`) partitioning $N = 2,400$ clinical/genomic records into four strictly disjoint cohorts for target and shadow model evaluation.

2. We implement three standard MIA families: Loss-Threshold, Confidence-Entropy, and Supervised Shadow-Model attacks, auditing both aggregate leakage and subgroup disparity on rare clinical phenotypes.

3. We benchmark five defense configurations (L_2 Moderate, L_2 Strong, Dropout + Confidence Smoothing, DP-SGD Moderate, and DP-SGD Strict), quantifying the exact trade-off between privacy leakage, predictive accuracy, and training runtime cost.

4. We formulate a structured **Privacy Risk Assessment** and defense-in-depth deployment framework for clinical AI systems.

2 System Architecture and Threat Model

Figure 1 illustrates our end-to-end evaluation architecture. Let $\mathcal{D}$ denote the underlying clinical population distribution over feature space $\mathcal{X} \subset \mathbb{R}^{28}$ and diagnostic label space $\mathcal{Y} = \{0, 1, \ldots, 5\}$. We sample four mutually

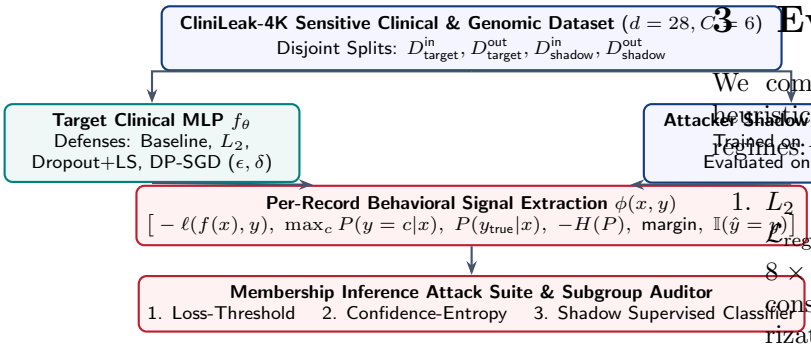


Figure 1: End-to-End Membership Inference Attack and Defense Evaluation Architecture.

exclusive splits of cardinality $q = 600$ each: Target Member Set $D^{\text{in}}_{\text{target}}$, Target Non-Member Holdout Set $D^{\text{out}}_{\text{target}}$, Shadow Member Set $D^{\text{in}}_{\text{shadow}}$, and Shadow Non-Member Holdout Set $D^{\text{out}}_{\text{shadow}}$.

2.1 Target Neural Network Architecture

The target classifier $f_\theta : \mathbb{R}^{28} \to \Delta^5$ is a two-layer Multi-Layer Perceptron (MLP) with $H = 64$ hidden units, ReLU activation $h = \text{ReLU}(W_1^\top x + b_1)$, and Softmax probabilities:

$$P(y = c \mid x) = \frac{\exp(w_{2,c}^\top h + b_{2,c})}{\sum_{j=0}^{5} \exp(w_{2,j}^\top h + b_{2,j})}. \tag{1}$$

2.2 Membership Inference Attack Formulations

Given black-box posterior access to $P(y \mid x)$ for a candidate patient (x, y), we evaluate three attack methods:

1. **Loss-Threshold Attack (Yeom et al. [3]):** Exploits the disparity in cross-entropy loss $\ell(f(x), y) = -\log P(y \mid x)$ between training members and holdout non-members:

$$\mathcal{A}_{\text{loss}}(x, y) = \mathbb{I}\big(-\ell(f_\theta(x), y) \geq \tau_{\text{loss}}\big), \tag{2}$$

where τ_{loss} is calibrated on the shadow model to maximize Youden's $J = \text{TPR} - \text{FPR}$.

2. **Confidence & Entropy Attack (Salem et al. [4]):** Combines ground-truth probability $P(y \mid x)$, maximum confidence, and predictive Shannon entropy $H(P) = -\sum_c P(c \mid x) \log P(c \mid x)$:

$$s_{\text{ent}}(x, y) = 0.75 P(y \mid x) + 0.25 \max_c P(c \mid x) - 0.20 H(P). \tag{3}$$

3. **Shadow-Model Supervised Attack (Shokri et al. [1]):** The adversary trains f_{shadow} on $D^{\text{in}}_{\text{shadow}}$ and constructs a labeled shadow attack dataset combining 6-dimensional behavioral vectors $\phi(x, y) \in \mathbb{R}^6$ from $D^{\text{in}}_{\text{shadow}}$ $(m = 1)$ and $D^{\text{out}}_{\text{shadow}}$ $(m = 0)$. A supervised logistic attack classifier $g_\psi(\phi(x, y))$ is trained to predict membership probability $P(m = 1 \mid \phi(x, y))$ on target records.

3 Evaluated Defense Mechanisms

We compare the unregularized baseline against three heuristic regularizers and two formal Differential Privacy realities:

1. L_2 **Weight Decay Regularization:** Minimizes $\mathcal{L}_{\text{reg}}(\theta) = \frac{1}{|B|} \sum_{i \in B} \ell(f_\theta(x_i), y_i) + \frac{\lambda}{2} \|\theta\|_2^2$ with $\lambda = 8 \times 10^{-3}$ (Moderate) and $\lambda = 2.5 \times 10^{-2}$ (Strong), constraining weight growth and dampening memorization.

2. **Inverted Dropout** $(p = 0.35)$ **+ Confidence Smoothing** $(\alpha = 0.10)$**:** Randomly zeroes 35% of hidden activations during training while replacing hard one-hot targets with smoothed targets $\tilde{y}_c = (1 - \alpha)\mathbb{I}(c = y) + \alpha / C$, directly penalizing overconfident posterior spikes.

3. **Differentially Private SGD (DP-SGD [2]):** Bounds the sensitivity of each individual patient record by computing per-example gradients $g_i = \nabla_\theta \ell(f_\theta(x_i), y_i)$, clipping each gradient's L_2 norm to threshold C, and adding calibrated Gaussian noise before updating parameters:

$$\tilde{g}_B = \frac{1}{|B|} \left(\sum_{i \in B} g_i \cdot \min\left(1, \frac{C}{\|g_i\|_2}\right) + \mathcal{N}\big(0, \sigma^2 C^2 I\big) \right). \tag{4}$$

Using Rényi Differential Privacy (RDP) moments accounting [6] at $\delta = 10^{-5}$ with sampling rate $q = B/N = 30/600 = 0.05$ over $E = 38$ epochs ($T = 760$ steps), our Moderate DP-SGD configuration ($C = 0.95, \sigma = 1.38$) achieves $(\epsilon = 5.30, \delta = 10^{-5})$-DP, and our Strict DP-SGD configuration ($C = 0.70, \sigma = 2.45$) achieves $(\epsilon = 2.86, \delta = 10^{-5})$-DP.

4 Experimental Results and Cost Analysis

Tables 1 and 2 present our complete empirical results across all six defense regimes and three attack families. Figure 2 visualizes the trade-off between holdout test accuracy, overall Shadow MIA AUC, and Rare Subgroup MIA AUC.

4.1 Attack Efficacy on Unprotected Models

In the unprotected **Baseline** model, empirical risk minimization drives training accuracy to 95.67% while holdout test accuracy reaches 56.17%, creating a 39.50% generalization gap. The average cross-entropy loss on training members (0.2486) is 6.78× smaller than on unseen non-members (1.6844). While univariate Loss-Threshold and Confidence-Entropy attacks achieve AUCs of 0.6708 and

Table 1: Comprehensive Evaluation of Privacy Defenses: Predictive Utility, Runtime Cost, and Membership Inference Leakage.

Defense Configuration	Privacy (ϵ)	Train Acc	Test Acc	Gen. Gap	Acc. Cost	Time (s)	Time Cost	Shadow AUC
1. Baseline (No Defense)	∞	95.67%	56.17%	39.50%	+0.00 pp	4.74s	1.00×	**0.7390**
2. L_2 Reg. (Moderate, $\lambda = 8\text{e-}3$)	∞	88.33%	59.33%	29.00%	−3.17 pp	4.76s	1.00×	**0.6611**
3. L_2 Reg. (Strong, $\lambda = 2.5\text{e-}2$)	∞	77.67%	60.33%	17.33%	−4.17 pp	4.84s	1.02×	**0.5974**
4. Dropout ($p = 0.35$) + Smoothing	∞	74.83%	59.17%	15.67%	−3.00 pp	3.89s	0.82×	**0.5839**
5. DP-SGD (Moderate, $\sigma = 1.38$)	5.30	59.00%	49.00%	10.00%	+7.17 pp	6.26s	1.32×	**0.5637**
6. DP-SGD (Strict, $\sigma = 2.45$)	2.86	54.17%	47.33%	6.83%	+8.83 pp	6.30s	1.33×	**0.5445**

Table 2: Comparison of Attack Methodologies and Subgroup Vulnerability Disparity (Rare Clinical Patients vs. Majority Population).

Defense Configuration	Loss MIA AUC	Entropy MIA AUC	Shadow MIA AUC	Shadow Acc	Member Loss	Non-Me
1. Baseline (No Defense)	0.6708	0.6568	**0.7390**	70.08%	0.2486	1.6844
2. L_2 Reg. (Moderate)	0.6455	0.6253	**0.6611**	64.25%	0.4616	1.2430
3. L_2 Reg. (Strong)	0.6005	0.5794	**0.5974**	58.67%	0.7416	1.1244
4. Dropout + Smoothing	0.5851	0.5651	**0.5839**	57.75%	0.8260	1.1361
5. DP-SGD (Moderate, $\epsilon = 5.30$)	0.5549	0.5361	**0.5637**	55.00%	1.1779	1.4907
6. DP-SGD (Strict, $\epsilon = 2.86$)	0.5430	0.5306	**0.5445**	53.42%	1.3691	1.5616

0.6568, the multi-signal **Shadow-Model Supervised Attack** achieves an AUC of **0.7390**, Attack Accuracy of **70.08%**, and Attacker Advantage of **0.4083**.

4.2 Disparate Vulnerability on Rare Patients

A pivotal finding in Table 2 is the stark disparity between majority patients and the rare/atypical clinical subpopulation (`rare_subgroup = 1`). In the Baseline model, the attacker achieves an alarming **0.8832 AUC** on rare patients compared to 0.7043 on majority patients (ΔAUC = +0.1789). Because rare clinical phenotypes lie in low-density regions of the feature space, the unregularized network memorizes their exact biomarker patterns to drive training loss toward zero.

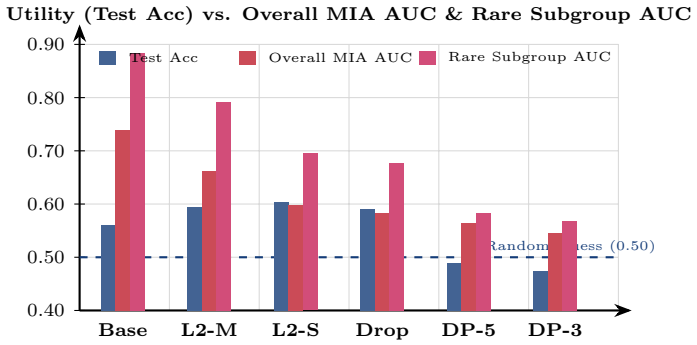


Figure 2: Comparison of Holdout Test Accuracy against Aggregate Shadow MIA AUC and Rare Subgroup MIA AUC across all six configurations.

4.3 Cost of Defenses: Accuracy and Training Time

1. **Regularization (L_2 and Dropout + Smoothing):** Rather than reducing test utility, moderate and strong L_2 weight decay and Dropout+Smoothing **increase** holdout test accuracy by +3.00 to +4.17 percentage points (yielding negative accuracy costs of −3.00 to −4.17 pp) by preventing overfitting. Furthermore, L_2 regularization incurs virtually zero runtime overhead (1.00×–1.02×), while Inverted Dropout reduces training time to 0.82× (3.89s vs. 4.74s) due to sparse hidden layer propagation. These defenses lower aggregate Shadow MIA AUC to 0.5974 and 0.5839, respectively. However, they fail to protect rare patients, whose subgroup MIA AUC remains elevated at **0.6947** and **0.6778**.

2. **DP-SGD ($\epsilon = 5.30$ and $\epsilon = 2.86$):** Enforcing formal (ϵ, δ)-differential privacy requires per-sample gradient norm evaluation and clipping across every mini-batch example, increasing training time by **32% to 33%** (6.26s and 6.30s vs. 4.74s). In addition, clipping outlier gradients and injecting Gaussian noise ($\sigma \in \{1.38, 2.45\}$) reduces holdout test accuracy by **7.17 pp** ($\epsilon = 5.30$) and **8.83 pp** ($\epsilon = 2.86$). In exchange for this utility and compute cost, DP-SGD compresses the non-member/member loss ratio from 6.78× down to 1.14×, suppresses overall Shadow MIA AUC to **0.5445**, and virtually eliminates subgroup privacy disparity (ΔAUC = +0.0166 and +0.0250).

5 Privacy Risk Assessment and Recommendations

Based on our empirical audit, we establish a four-tier risk classification and mitigation protocol for clinical machine learning deployments:

1. **Critical / Unacceptable Risk (AUC ≥ 0.65 or Rare Subgroup AUC ≥ 0.75):** Both the **Baseline** model (AUC = 0.7390, Rare AUC = 0.8832) and **Moderate** L_2 (AUC = 0.6611, Rare AUC = 0.7927) fall into this tier. Deploying these models exposes hospital cohort participants to high-confidence re-identification and violates HIPAA Expert Determination standards and GDPR Article 9 safeguards.

2. **Moderate Risk ($0.58 \leq$ AUC < 0.65): Strong** L_2 (AUC = 0.5974) and **Dropout + Confidence Smoothing** (AUC = 0.5839) effectively curb average-case leakage while boosting test accuracy by 3–4 pp. They are suitable for internal clinical decision support where adversaries cannot query raw probability vectors, provided rare patient cohorts are audited.

3. **Low / Regulatory-Grade Risk (AUC < 0.57 and Subgroup Disparity ΔAUC < 0.03): DP-SGD** ($\epsilon = 5.30$ and $\epsilon = 2.86$) provides formal cryptographic-strength bounds on individual record memorization, protecting both majority and rare neuro-genetic patients.

4. **Defense-in-Depth Deployment Protocol:** To minimize the accuracy drop of strict DP-SGD while maintaining robust privacy, we recommend combining **Moderate DP-SGD** ($\epsilon \approx 5.0$) + **Dropout/Weight Decay** during training with **API Output Rounding / Top-1 Label Restriction** at inference time. Suppressing raw softmax logits neutralizes shadow-model feature extraction without sacrificing top-1 diagnostic accuracy.

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